軟體測試

期末考

考試中，請勿參閱資料。

1. Please explain Beizer testing level 0, 1, 2, 3, and 4.　Please also give examples in practices for their differences.
   Level 0: (2pts/2)

   Level 1: (2pts/4)

   Level 2: (2pts/6)

   Level 3:(2pts/8)

   Level 4:(2pts/10)

2. Below is a graph, defined by the sets of nodes, initial nodes, final nodes, edges, and defs and uses. Each graph also contains a collection of test paths. Answer the following questions about each graph.

Graph: N={0,1,2,3,4,5,6,7}

N0={0}

Nf={7}

E={(0,1),(1,2),(2,7),(2,3),(2,4),(3,4),(4,5),(5,6),(4,6),(6,2)}

def(0)=def(5)=use(4)=use(2)={x}

Test paths:

t0=[0,1,2,7]

t1=[0,1,2,4,6,2,7]

t2=[0,1,2,4,5,6,2,7]

t3=[0,1,2,3,4,6,2,7]

t4=[0,1,2,3,4,5,6,2,7]

t5=[0,1,2,3,4,5,6,2,4,5,6,2,7]

(a) Draw the graph. (5pts/15)

(b) List all of the du-paths with respect to x. (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/20)

(c) For each test path, determine which du-paths that test path tours. For this part of the exercise, you should consider both direct touring and sidetrips. Hint: A table is a convenient format for describing this relationship. (5pts/25)

(d) List a minimal test set that satisfies all-defs coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/30)

(e) List a minimal test set that satisfies all-uses coverage with respect to x. (Direct tours only.) Use the given test paths. (6pts/35)

(f) List a minimal test set that satisfies all-du-paths coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/40)

3. Use the following predicate to answer the following questions.

$$p = \neg(a \vee (\neg a \wedge b) \vee (\neg b \wedge \neg c))$$

(a) Identify the clauses that determine predicate p. (5pts/45)

(b) Compute (and simplify) the conditions under which each of the clauses determines predicate p. (5pts/50)

(c) Write the complete truth table for all clauses. Label your rows starting from 1. Use the format that row 1 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (5pts/55)

(d) Identify all pairs of rows from your table that satisfy general active clause coverage (GACC) with respect to each clause. (5pts/60)

(e) Identify all pairs of rows from your table that satisfy correlated active clause coverage (CACC) with respect to each clause. (5pts/65)

(f) Identify all pairs of rows from your table that satisfy restricted active clause coverage (RACC) with respect to each clause. (5pts/70)

(g) Identify all 3-tuples of rows from your table that satisfy general inactive clause coverage (GICC) with respect to each clause. Identify any infeasible GICC test requirements. (5pts/75)

(h) Identify all 3-tuples of rows from your table that satisfy restricted inactive clause coverage (RICC) with respect to each clause. Identify any infeasible RICC test requirements. (5pts/80)

4. We have a program that accepts three integer input parameter a, b, and c. The progam then solves ax2+bx+c =0 and returns the solution if any. Please show how to apply the input space partitioning technique to design test input for completeness and disjointness among blocks. (10pts/90)

5. Assume that we have the following program with two mutants.

Original program:

```
public static int count(int[] x, int y) {
  int c = 0;
  for (int j = 0; j < x.length; j++)
    if (x[j] == y) c++;
  return c;
}
```

Mutant 1:

```
public static int count(int[] x, int y) {
  for (int j = 0; j < x.length; j++)
    if (x[j] == y) c++;
  return c;
}
```

Mutant 2:

```
public static int count(int[] x, int y) {
  int c = 0;
  for (int j = 0; j <= x.length; j++)
    if (x[j] == y) c++;
  return c;
}
```

(a) Please write down a test input that kills mutant 1. Explain why it kills the mutant? (5pts/95)

(b) Please write down a test input that kills mutant 2! Please explain why it kills the mutant. (5pts/100)